

CVP Capstone Project Status Report*

CVP 613

Capstone Project Name: CPB Support Tool
Project Managers (Group): Sydney Lias and My Do
Date of Status Entry: 8/5/26
Period Covered: 6/25/26 - 8/5/26
Project Start Date: 4/25/26

Summary and Overall Project Status

Enter information about the overall project health status and highlights

Overall Status: GREEN (On Track)

The CPB Support Tool remains on track and has progressed since the previous status report. The working prototype now includes perfusion, prime, anticoagulation, cannula selection, drug library, electrolytes/blood gas, blood products/blood management, and master references sections. The team has also aligned the project with updated Proposal 1 and Proposal 2 language, including the pilot-study design, planned outcomes, data-management approach, human-subjects protections, final presentation, and poster-support materials.

No critical red items are identified at this time. The main attention areas are IRB approval/determination, final participant-facing materials, simulation dry-run preparation, source verification, and keeping the Fall 2026 pilot scope aligned with approved introductory-student tasks. Budget impact remains minimal; the main constraints are time, IRB readiness, source verification, and keeping tool development synchronized with capstone documentation.

Milestones/Tasks

Not Yet Started	On Track/In Progress	At Risk	Off Track	Completed	Responsible Team Member
Weekly project coordination and milestone tracking - GREEN					My/Sydney
Core CPB calculator build (perfusion, prime, anticoagulation) - BLUE					Team
Anticoagulation expansion: heparin tally, HDR curve, AT3, bivalirudin - BLUE					Team
Cannula selection workflow and pressure-drop visualization - GREEN					Team
Drug, electrolyte/blood gas, and blood product reference modules - GREEN					Team
Master references optimization and module-based source organization - BLUE					Team
Proposal 1 and 2 content integrated into project notes and presentation - BLUE					Team
Final capstone presentation and poster-support materials - BLUE					Team
Participant-facing IRB materials and data-collection tools - YELLOW					Team/PI
IRB approval/determination before recruitment or data collection - YELLOW					PI/Team
Fall 2026 pilot simulation preparation and case rehearsal - GREEN					Team
Pediatric mode and additional future drug-library expansion - GREY					Future scope

Highlights and Key Takeaways

Bullet points of great work, who owns what tasks, where teams are pivoting, etc.

Working prototype has expanded beyond the June build and now covers calculation support, workflow visualization, and reference/learning support.

Anticoagulation support now includes heparin loading, additional heparin projection, heparin administration log, protamine from tallied heparin, AT3 dosing, bivalirudin loading dose, and a heparin response curve.

Cannula selection now includes arterial, venous, and optional bicaval planning with flow-based pressure-drop visualization.

Reference content has expanded to include a drug library, electrolyte/blood gas ranges, blood products and blood management, and a module-organized master reference page.

Proposal 1 and Proposal 2 content has been incorporated into the current project identity, methods, outcomes, data plan, human-subjects guardrails, and presentation/poster materials.

Next steps are to complete IRB readiness items, rehearse the simulation case and scoring process, and keep the Fall 2026 pilot scope aligned with the approved protocol.

*Adapted from smartsheet.com and LinkedIn

Project Timeline GANTT Chart

Examples of Tasks & Milestones	6/25	7/4	8/1	8/15	9/1	10/1	11/1	12/1
Regular project team meetings								
Proposal drafts/updates								
Proposal / presentation drafts and final								
IRB determination / approval								
Update ROL / study flow diagram								
Create/update evidence table								
Update literature appraisal/synthesis								
Conduct organizational assessment								
Update methods and data plan								
Data management and security plan								
Communication plan / recruitment materials								
Budget/constraints monitoring								
Workflow diagram / implementation plan								
Simulation dry run and pilot prep								

Recruitment after IRB approval								
Simulation pilot sessions								
Data review and tool refinement								
Final report / dissemination								

Updated Action Items

Confirm formal IRB approval or determination before recruitment or pilot data collection.
 Finalize consent, recruitment, simulation instructions, surveys, debrief questions, checklist, timing method, and scoring guide.
 Run a simulation dry run to test case flow, student instructions, tool access, timing capture, and data collection.
 Freeze the approved pilot feature scope and keep pediatric mode or additional drug-library expansion in future scope unless approved.
 Continue documenting verified formulas, source references, and calculation test cases.

*Updated from the 6/25/26 version using current project notes, proposal-integrated planning language, README status, and successful verification run on 8/5/26.